AGRICULTURAL GLOCALIZATION: SYSTEM DEVELOPMENT OF MARKET MOBILE APPLICATION FOR SUSTAINABLE LOCAL INDUSTRY IN THE PHILIPPINES

Conference Paper · September 2020

DOI: 10.13140/RG.2.2.16684.90246

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Abstract

Natural resources are the source of all products produced and manufactured around the world. It is a fundamental component of all secondary products, such as food, clothing, shelter, automobiles, and others. The Philippines is rich in mineral and natural resources, including a wide variety of locally extracted raw materials. These materials are obtained and collected from the natural environment by the primary sector, for human consumption and use, and as a component for converting them into new products, either for sale or for export by the secondary sector; while the tertiary sector provides services to the public. The role of the primary sector is crucial for the sustainability of the local industry. They are the bedrock of social and economic development that contributes to the Philippine economy.

While the primary sector plays a major role in the country's economic growth, the 2020 Report survey conducted by the Philippine Statistics Authority (PSA), the primary sectors are one of the poorest of the poor. The largest incidence of poverty among the core sectors was recorded in 2018 in 31.6%, 26.2%, 24.5%, and 23.9% of farmers, fishermen, rural residents, and children belonging to families below the official poverty rate, respectively. These sectors also recorded 40.8%, 36.9%, 34.0%, and 33.5% as the highest levels of poverty in 2015 (PSA 2020).

This study was conducted by the researchers to help the primary sector to improve their economic activity, and to have a stable income through the creation of the Market Mobile Application. Application software for mobile devices such as mobile phones, smartphones, laptops, personal digital assistants (PDAs), tablets, or any other devices that can directly connect the primary sector (seller) to people (buyers) anytime and anywhere. It will provide the primary sector with new technologies to improve their livelihoods and develop their knowledge and skills using mobile technology as a platform to increase their productive activities, obtain information on daily weather conditions, and communication channels. Eventually, this innovation would turn them into entrepreneurs and business owners. This research will also highlight the economic activity involved in a sales transaction of the Primary Sector-to-Buyer (Ps2B) through a mobile application.

A mixed research methodology for quantitative and qualitative data was used in this study. A structured survey questionnaire in the Four Point Likert Scale was distributed to the respondents. To support the results of the survey, the researchers conducted a scoping review of other relevant studies. Both results were synthesized and analyzed using the Triangulation Analysis.

Keywords: *Local Industries, Mobile Application, Sustainable, Glocalization, Primary Sector*

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OBJECTIVES OF THE STUDY

To provide the primary sector a tool/instrument not only for business but also for access to government resources, projects and programs, occupational legislation, daily weather alerts and forecasts, interactive chat rooms, export opportunities, and best practice data, anywhere at any time.

To offer the primary sector an efficient platform to sell their products, which will eventually help them expand their economic operation and provide steady income through the development of the Market Mobile Application.

To know the role of the Public Administration in the development of the Market Mobile Application.

INTRODUCTION

The Philippines is an archipelago country of 7,107 islands covering 300,000 square kilometers (USAID, 2011) or approximately 30 million hectares (FMB, 2017). It is rich in natural and mineral resources (ECCP, 2019), a valuable asset that contributes to the country's national wealth, and a major source of income and livelihood for the people of the Philippines. The continuous improvement of these resources is significant for the country's Gross Domestic Product (GDP), and its economic growth.

Three main local industries are contributing to the country's economic growth. The primary sector, secondary sector, and the tertiary sector. The primary sector extracts its products/goods from nature such as plants, animals, trees, fish, minerals, and other natural resources. They produce raw materials from natural resources. The secondary sector, on the other hand, produces its products by converting raw materials into new products, such as trees into papers, fruits into preserved fruits, and meat into processed foods, among others. While the tertiary sector provides services to its stakeholders such as call center agents, teachers, and more.

The primary sector is the focus of this study. This sector is directly dependent on the environment/natural resources, such as land, forests, water, plants, and minerals (NSCB, 1998). It supplies raw materials to the secondary sector who convert it to another product, for human use and consumption. They are also called the "extractive sector", engaged in agricultural and harvesting activities, crop production, fisheries, forestry, hunting, livestock breeding, and mining; or "red-collar workers" because of the outdoor aspect of their work and activities. (George, 2020).

The products sold by the primary sector are naturally supplied. Abundance in nature is presumed to be sufficient to ensure them a sustainable livelihood and a comfortable life. However, the researchers found, that the primary sector is among the poorest of the poor (PSA, 2020). The situation is further compounded by the younger generation, who mostly decides to leave rural areas due to poverty, resulting in the depletion of several potential domestic workers or primary sectors.

The welfare of primary sectors, such as farmers, fishermen, craftsmen, animal breeders, and miners in developing countries, depends heavily on the selling price of their products or the value of raw materials sold (Jensen, 2007). The predominant exchange and market conditions between the primary sectors and traders are highly disadvantageous to the interests of the primary sectors, as the raw materials purchased by traders from them are too cheap and then sold to buyers at a higher price.

However, if the raw materials are sold directly from the primary sector to buyers (Ps2B), the decision to determine the price belongs to the primary sector, either by selling the products at the current market price or a lower price. This would make it possible for the primary sector to benefit more and to have a steady source of income since they can sell their products either to traders or directly to buyers.

For whom is the Market Mobile Application?

Market Mobile App's key stakeholders are the public administration, which will regulate the operation, provide training opportunities for the primary sector, set up government programs, provide export opportunities, and bridge partnerships with telecommunications companies for possible Wi-Fi-mobile device package; and with transport groups for the delivery of purchased goods. Secondly, the primary sector who extracts raw materials from natural resources; and, finally, the buyer who purchases local products. They are consumers, customers, traders, merchants, and more.

Theoretical Framework

The researchers in the conduct of this study have adopted the Expectancy-Value Theory of Motivation (EVTM) developed by Eccless and Wigfield (2002) and the Digital-Divide Framework created by Lloyd Morrisett (1995).

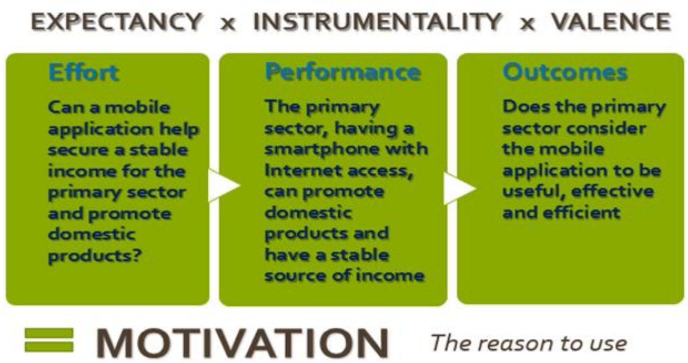


Figure 1: Expectancy-Value Theory of Motivation by Eccless and Wigfield (2002)

The EVTM postulates that motivation for a particular potential outcome is measured in two ways: 1) Expectancy, the probability to obtain the person's desired (instrumental) outcome based on actions/decisions; and 2) Value, the ideal outcome expected by the person as a result (Eccles & Wigfield, 2002). These factors are associated with multiplication, for example, motivation = expectation x value. Both expectations and values under this theory have a significant impact on the determination of future actions, commitment, resilience, and individual success.

Figure 1 shows that the primary sector expects the Market Mobile Application to provide them with a platform for advertising domestic products; a source of information on news, weather conditions, shared best practices, and government initiatives; a means of communication; and a venue for training opportunities to improve productivity. They also expect it to be user-friendly with Wi-Fi access. In the same way, the buyer expects the application to be convenient and user-friendly, that brings benefit and good experience in using it. Exceeding users' expectations of using the mobile app will generally result in positive response and value.

"Digital divide" was presented in the mid-1990s and described as the disparity between those who "have access" to the latest information, system tools, and technology and those who "do not have access" (Van Dijk, 2006). It remains controversial in public policy on political, social, and economic issues (Srinuan et al., 2011). In this research, this is illustrated by the primary sectors living in remote areas where access to the internet is usually impossible, but with the emerging mobile technologies, they have been able to use mobile phones to communicate with friends and their loved ones who are far from them.

With this, the Market Mobile application will not only provide the primary sector with a mode of communication but also facilitate a stable source of livelihood.

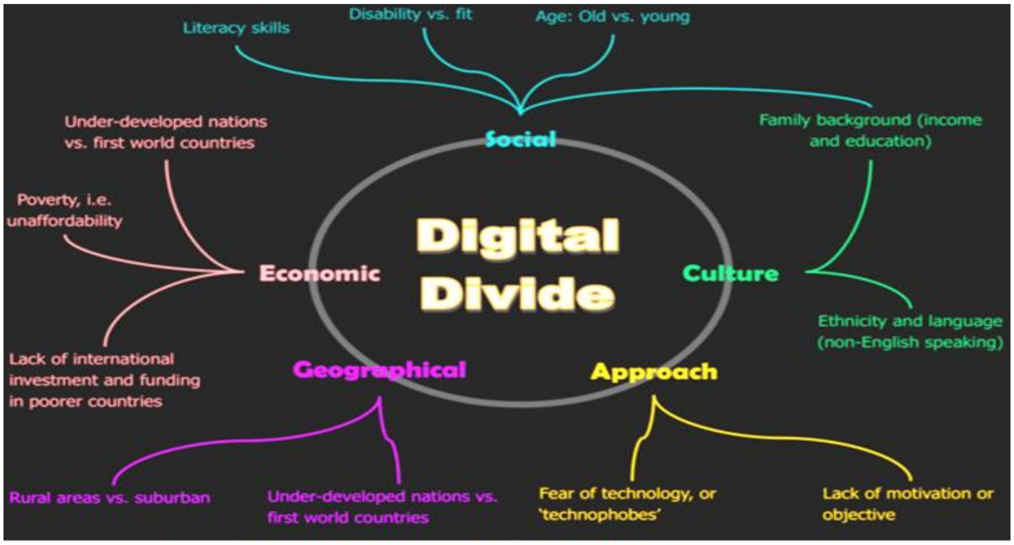


Figure 2. Digital Divide Framework introduced by Lloyd Morrisett (1995)

Source: Barupal, Shweta (2017)

In Figure 2, Barupal (2017) shows the digital divide's resulting factors such as social, culture, approach, geographical, and economic. The **Social factor** pertains to one's literacy skills and knowledge or technology know-how; disability and fitness which refers to the mobility of the user to use a mobile application; and, the young generation and the old users which relates to the exposure to technology. The difference between all these sub-factors is the perceived gap in the extent of the user's ability to use mobile technology, resulting in a digital divide. **Culture** covers the family background (income and education), ethnicity, and language. This gap relates to external factors beyond the control of the user, such as the exclusivity of a mobile application to a country using its language and not being translated into English; or the inadequate family income to purchase a smartphone that is compatible with a mobile application or do not have an Internet/Wi-Fi access. As for the **Approach aspect**, fear of technology or lack of motivation and the objective of using mobile technology. This refers to the effect of one's reaction and the acceptance/rejection of modern technology. As to **Geographical factor**, it distinguishes rural areas from suburban areas; and underdeveloped nations from first world countries. This includes the availability of internet access, the conventional or advanced technology of one country compared to another, and the type of smartphone available in a country that is compatible with a mobile application. And lastly, as to **Economic factor**, it determines whether a nation is under-developed or a first world country, have poverty issues and/or has a lack of international investments and funding in poorer countries. This concerns the economic initiatives of the public administration and its sound financial system to support the user's needs.

From traditional business transactions to mobile business apps, the researchers developed an innovative mobile technology for e-commerce, with products being sold through a mobile device using apps. This will reduce the sales and distribution cycle. The primary sector can sell its products directly to buyers (Ps2B), bridging the gap between people living in rural provinces with urban cities and other areas across the country.

Glocalization of Mobile Technology

The Mobile Market application can provide a platform and an opportunity to respond to different buyers' demands for basic food and commodities. The emergence of mobile technology and global opportunities tailored to meet local needs and interests is referred to as the "glocalization of mobile technology."

Figure 3: Glocalization of Mobile Application

Glocalization refers to the balance of global and local communities with a global community network designed to strengthen the social, cultural, and economic sectors through collaborative efforts in infrastructure and reform initiatives (Glocal Forum, nd). In short, "think global, act locally" (Brillantes and Perante-Calina, 2018).

The framework illustrated in Figure 3 was used to develop the Market Mobile Application. It shows Public Administration's response to technology with global perspectives, approaches, and opportunities to develop mobile applications that offer local and export opportunities, partnerships and ultimately improve the socio-economic situation of the country by promoting local industry and strengthening local products.

The Public Administration plays a significant role in making this progress a reality by designing strategies to boost market income for the primary sector, giving its children opportunities to sell their products, and enabling them to participate in partnership or collaboration with the private sector, civil society, and international investors. The private sector, civil society, international investors, and public administration should therefore respond to the challenges of globalization in the context of their systems, environments, and perspectives. With their respective jurisdictions, specific responses and impacts of mobile technology can be discussed and filtered in the same way as meeting the demands of the emerging global technologies.

In Figure 4, Van Dijk (2006) illustrates the cumulative and recursive approach of successive types of access to digital technologies. First, the user must be motivated to use mobile technology as a driving force in assessing its efficiency and effectiveness. Second, there should be available mobile devices compatible with a mobile application with access to the Internet, referred to as 'material access.' Third, the user must have the necessary skills (access) to use the mobile application as well as the competencies to explore the features of the mobile devices. Fourth, the user's ability to explore the features of a mobile application to validate its functionality to operate is referred to as "usage access." However, if the mobile device does not have full access to the mobile application, the user can go back to "material access" to evaluate the usability of the application or upgrade the mobile phone. The cycle will continue until the next innovation of mobile devices is made to meet the changing needs of the user. To avoid the gap caused by the digital divide, these factors should be considered by the public administration.

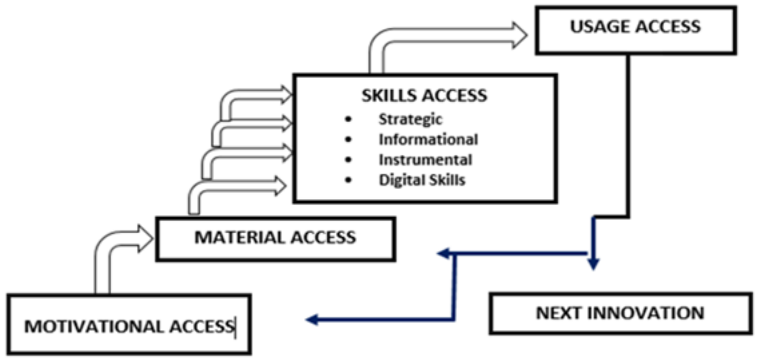


Figure 4. A Cumulative and Recursive Method of Successive Kinds of Access to Digital Technologies

Source: van Dijk (2005)

Van Dijk (2006) also discussed that a person's first wants or desires (motivation) to have a computer and to be connected to the Internet before having physical access to the computer. With this, digital technology appears to be not only "have-nots," but also "want-nots." This explains that the primary sector may want to have (want-nots) a smartphone. However, this desire will remain a desire due to poverty. Instead of buying (have-not) a smartphone, they need to feed their family first; and despite the potential change that they might have in the future, their focus is on their current situation.

Conceptual Framework

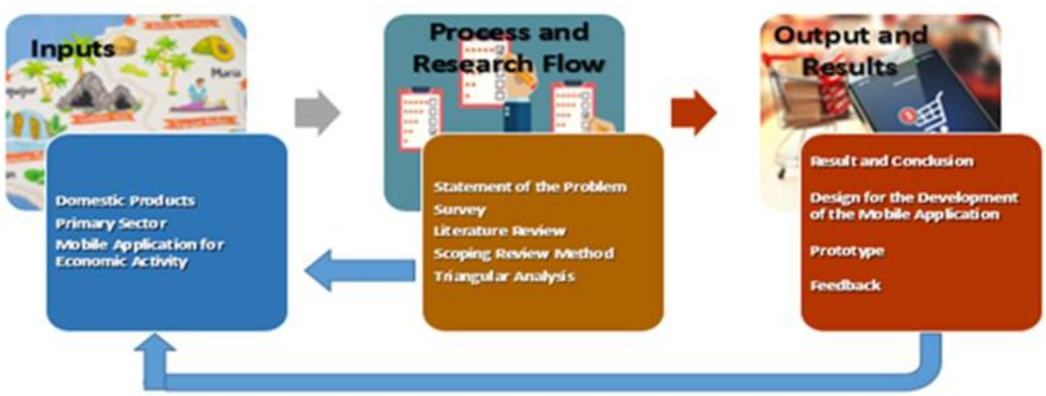


Figure 5. Conceptual framework of the study

The researchers have been motivated to undertake this research to provide the primary sector with an alternative means of selling their products and increasing their profits and to offer additional options to have a stable and secure income. Many young people are passionate about using mobile apps and technologies. This enthusiasm will also make it possible for the primary sector to motivate their children to embrace their industry and encourage them to continue advocating the sale of domestic products.

The researchers used the structured questionnaire using the Four-Point Likert Scale to gather information and data to address the statement of the problem. To further support the findings of the survey, the researchers conducted a scoping review to identify gaps in mobile applications. Lastly, to synthesize both findings in the survey and the scoping review, the Triangular Analysis was conducted.

The development of the Market Mobile application was designed based on the outcome of the Triangular analysis (Figure 5).

There are 284 primary sector and 275 buyers as respondents of the study. They are randomly selected using an e-mail address, Facebook, and messenger. Most parts of the Philippines were placed under community quarantine due to coronavirus. The researchers, therefore, recognize that there is no equal representation of the primary sector (subsectors) due to mobility restrictions.

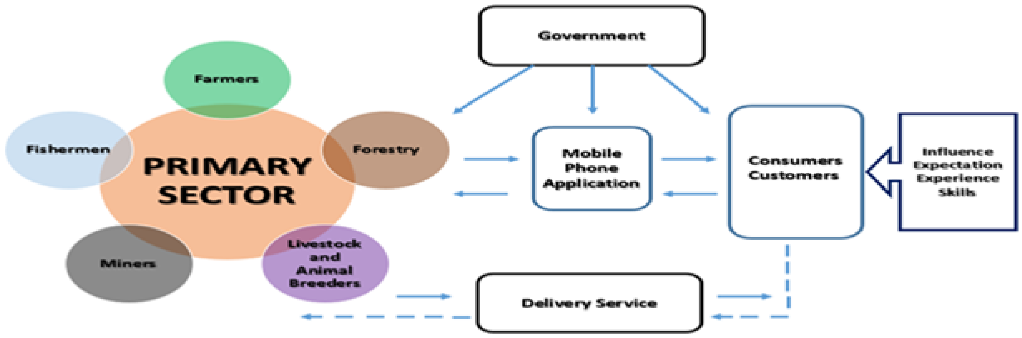


Figure 6. Market Mobile Application Framework

The study concluded with the creation of Market Mobile Application's Framework and its prototype that can be used by the primary sector (Figure 6) to market their domestic products and other relevant activities. This software is a user-friendly application that offers flexibility and typically handles users' uniform needs, regardless of the variety of products sold, and can often adapt to the unique capabilities and limitations of different mobile devices (Taylor and Levin, 2014). Buyers expect a mobile application that will provide distinct advantages, rather than simply function as virtual "repackaged" websites across mobile platforms (Newman, et al. 2018). The market activities of the Market Mobile application process, as shown in Figure 6, developed by the researchers, has three main players: The Primary Sector, the Buyer, and the Public Administration. The support groups to complete the market activity of this mobile application are the payment centers, payment platforms, and transport services for the shipping of the purchased products.

STATEMENT OF THE PROBLEM

This research was conducted to answer the question: In what way can the primary sector improve their livelihood and market activities?

The research questions are stated, as follows:

1) What is the assessment by the Respondents of their accessibility in terms of:
   a. Mobile devices
   b. Internet
   c. Delivery services
   d. Payment centers

2) How can a mobile application affect the livelihoods of the primary sector?

3) What is the impact on buyers using a mobile application?

REVIEW OF RELATED LITERATURE

Mobile Device Application

Mobile technology is a fast-growing field that is closely linked to individual work and daily activities. Continuous innovations with positive and negative effects contribute to the development of new methods of application that are expected to help the user (Jayatilleke, et al., 2018).

"Mobile device" refers to a range of devices that allow users from elsewhere to access information and messages (Karetsos et al., 2014). It is a compact, lightweight gadgets such as smartphones, mobile phones or handsets, laptops, and iPads (Jayatilleke, et al, 2018) that can be used anywhere and anytime. Mobile devices have become an important partner of an individual, helping them carry out their activities, practices, and transactions. It has been an important tool to accomplish something.

People install various applications in their smartphones and other devices to suit their needs and preferences. Mobile phones have influenced many areas along with electronic networking systems (Traxler, 2009), and people's day-to-day activities, including shopping, recreation, careers, and one-to-one social experiences. Mobile technology with a useful software application increases competition improves product performance, reduces risks, and protects the environment while at the same time increases profitability, revenue, economic growth, and sustainability.

Since mobile phones are the first digital communications handset technology for many urban residents, the primary sectors are extremely worried that the transition to advance digital technology may lead rural communities to struggle more to see how and when mobile devices can be used productively (FAO, 2013). Mobile phones are the most effective tool for rural communications and key innovations to facilitate social change and improve local industries (Hernandez, 2012). It also provides tremendous opportunities for direct linkages to the primary sector through extension programs, professionals, suppliers, and buyer knowledge tools. Mobile phones also may offer a convenient tool for an emergency, weather, and business communication (Nyiri, 2005).

According to Aker (2011), SMS text messages are used because they are easy to create, customize, and cost-effective for a wider selection of communication. Mobile phones can be a tool for on-site contact with buyers (Chhachhar, 2012). However, irrespective of the extent to which it is used, there are also limitations on each mobile communication platform. This includes a high level of illiteracy that requires real-time interaction with the primary sector (Duncombe, 2012).

Most rural people use conventional mobile phones with a minimal alternative mobile interface (Chisama, 2016). No matter how noble the purpose of developing a mobile application for the primary sector, this initiative would be useless if they only had a conventional mobile phone and/or no internet access. It is therefore important that their situation should be considered, and the necessary reliable information is supplied before creating a mobile app. The lack of "material access" for use including mobile phones and the Internet prevents the primary sector from contributing more to the Philippine economy. The public administration should, therefore, intervene and bridge the gap created by the "digital divide" to the primary sector.

Mobile applications should provide users with access to the services and products offered by the seller to suit their needs in a variety of ways (Wang et al., 2015). These relationships must be improved by ensuring that their applications are convenient and easy to use as possible (i.e. ease of use would make it easier for buyers to integrate technology into their concepts) (Newman, et al., 2018). As a result, users instinctively use mobile device applications to express and convey their desires, expectations, and wishes – virtually anytime and anywhere, simply by clicking the mobile phone button to access the application on their device (Belk, 2013).

M-commerce is identified as being convenient and ubiquitous (Shan Du and Hua Li, 2019). The three key aspects for m-commerce transactions are (1) security and trust; (2) personalization and localization; and (3) convenience of the buyer. Security and trust are the two main drivers for the growing demand for m-commerce in a country (Chantzaras, et al. 2017). The data of the buyer should be protected, particularly, if they use their credit card for payment and provide their address for delivery. The buyers, therefore, buy goods in a safe online environment. Personalization requires frequent contact with the buyers by encouraging them to update their profile such as a username (real name if possible), how often the buyer shops, or buyer's last transactions, and how long the buyer browsed online (Cash-Crowley, 2015). In terms of Localization, the seller must provide the buyer with the appropriate message and select the target buyer at the correct time and place. Information such as backgrounds, weather indicators, and community preferences can all be applied to a simple personalization and localization to encourage the buyer to continuously transact with the seller (id).

With the emerging mobile networks, digital marketing is easier to manage and cost-effective (Balasubramanian, et al., 2002). Many studies in the past focused on the benefits of mobile commerce, and this provides a pillar for further studies, as shown in Table 1.

Title of the Study	Author/s and Year	Theories/Framework	Brief Result of the Study
E-commerce Smartphone Application	Abdullah Saleh Alqahtani and Robert Goodwin (2012)	Technology Acceptance Model, Theory of Planned Behavior, and Theory of Reasoned Action	Perceived of usefulness, perceived ease of use, attitude, security, trust, subject norm and risk are possible factors that may affect online purchase intention.
Mobile Commerce (M-Commerce) Interface Design: A Review of Literature	Zaifulasraf Ahmad and Roslina Ibrahim (2017)	Two design concept which is dynamic and design concept; eight design guidelines that involve the page format design, flat structure, navigation, button, history list signal, remembering item and search query; and catalogue and payment site	Each mobile OS has its own design and limitations that can affect the design of m-commerce.
Research and Analysis of the Development Approaches in E-Business Mobile Application	Yongkang Xing (2019)	Phone Gap Application working principle	The development in Phone Gap is based on a hybrid development environment to build the application in HTML 5 rendering engine with Phone Gap plug-ins to access mobile API. The hybrid working principle allows Phone Gap to access most of the mobile API.
Comparative Study of Online Shopping Experience With Specific Reference to Mobile Apps Based Shopping	Madhavi Damle, Avinash Aslekar, and Subrahmanyam Yadavalli (2016)	Internet shopping and traditional brick and mortar store shopping.	Companies cannot do away with either apps or web based app both are going to stay. Companies need to formulate appropriate strategy having a balanced mix of app and mobile web.
Mobile Application Development Approaches: Recommendation for E-commerce Enterprises	Mahmudul Hasan, and Ariful Haque (2016)	Cordova Framework Tool Selection Matrix	In spite of the apparently rational outcome, the "tool selection matrix" might have some limitations. Therefore, its efficacy and effectiveness of implication as well as possible limitations could be subject to further research.
Mobile Application and Its Global Impact	Rashedul Islam, Rofiqul Islam, Tohidul Arafhin Mazumder (2010)	Mobile application	The modern mobile applications are more capable and more usable for the user. And the global impacts of mobile applications are going high
Search Convenience and Access Convenience: The Difference Between Website Shopping and Mobile Shopping	Ghaith M. Jaradat, Ibrahim Almarashdeh, Mutasem Alsmadi (2019)	Website Shopping and Mobile Shopping Access convenience and Search convenience	The results show that mobile apps is more adopted in term of accessibility but website shopping is more adopted in term searchability. This study did not find significant difference it term of total adoption of both website and mobile shopping.
Evolution of the Digital Divide in Europe	Carmen Boje, Gary Steffen, and Nicolae-George Dragulanescu (2018)	Digital divide	To fill the gap created by the Digital Divide at local, national or international levels does not just mean providing computers to people who need them. It also means that people need to be trained in how to use them, and more importantly, they must understand how to access and use the information. Access not only includes knowing where and how to locate information, but also how to understand it, and how to use it wisely.
Moderating effect of the digital divide of e-commerce	Javier A. Sánchez-Torres (2019)	Technology Adoption Model Digital Divide Electronic Commerce	The explanatory result validates that the perception of the digital divide negatively moderates the influence of the variables related to internet technology on the use of e-commerce.
E-Commerce and Digital Divide: Impact on Consumers	T.P. Rama Rao (2001)	Digital Divide E-Commerce Capacity Building	Rapid developments in ICT have opened up new global business opportunities in the form of E-Commerce, which may exploited by the developing countries as well. It is feared that these developments may widen the digital divide and under developed countries may lag behind and lose in the race
Global E-Commerce: Rationale, Digital Divide, and Strategies To Bridge The Divide	Lakshmi S Iyer, Larry Taube, and Julia Raquet. (2014)	Macro-Economic Perspective Pyramid of E-Commerce Strategies Digital Divide	Although globalization is regarded as a phenomenon that brings various advantages to the worldwide economy, based on the extensive literature review and data obtained from secondary sources, the authors find significant differences in the growth of global EC
The Difference Between Shopping Online Using Mobile Apps and Website Shopping: A Case Study of Service Convenience	Ibrahim Almarashdeh, Ghaith Jaradat, Anmar Abuhamdah, Mutasem Alsmadi, Malik B Alazzam. Raed Alkhasawneh, and Ismaeil Awawdeh (2019)	Technology Acceptance Model Unified Theory of Acceptance and Use of Technology	The results of data analysis illustrated that the customers feels more comfortable using mobile apps for online shopping than online website in term of search convenience, access convenience and service recovery convenience.
Expectancy–Value Theory of Achievement Motivation	Allan Wigfield (2000)	Expectancy–Value Model	These results provide support for some of the proposed links in the Eccles et al. (1983) model of achievement choice; in particular, the links between previous performance and subsequent ability beliefs and ability belief and subsequent performance
Social Psychological Determinants of Mobile Communication Technology Use and Adoption	Oscar Peters (2007)	Gratifications Approach Expectancy-Value Theory Social Cognitive Theory Unified Theory of Acceptance and Use of Technology Theory of Reasoned Action Theory of Planned Behavior	The expectancy-value judgments model of uses and gratifications should perform well in explaining existing media technology behavior in terms of beliefs and expectations. The unified model of acceptance and use of technology should perform well in supplying general information on user's opinions about a media technological innovation in terms of expectancies and conditions of use.
Poverty and Agriculture in the Philippines: Trends in income poverty and distribution	Celia M. Reyes, Aubrey D. Tabuga, Ronina D. Asis, and Maria Blesila G. Datu (2012)	Subsistence Incidence Income inequality	To illustrate, almost 7 out of 10 poor workers (68%) in 2009-2010 who are underemployed are primarily engaged in agriculture, forestry, or fishery

Table 1. Scoping Review on Different Studies for the Development of Market Mobile Application

Digital commerce has now appeared in several mobile applications and website networks. It provides other functions and services, such as payment, advertising, online ordering, online transactions, and real-time headlines (Du and Li, 2019). Developing a mobile application enables developers to design and create a mobile app that responds to users' needs and guarantees protection of information, taking account of the possibility of data loss or unauthorized use of their account.

7Cs of Interface Implementation	M-Commerce	
	Mobile Setting *To support consumers; limited attention*	**Mobile Device Constraints** *To complement the insufficient display of mobile devices*
Context	Menu structured in a shallow rather than a deep hierarchy Layered sequential process rather than field selection process	Summary and keywords that give a whole picture of information separated over pages.
Content	Proximate selection method that makes nearby located-objects easier to choose (gas stations, bank accounts)	Conversion of visual information to audio format Use of non-speech sound
Community	Connection to shopping companions who share interests in common	SMS, and graphics describing products, transferred through a user's phone book
Customization	Proximate selection method that emphasizes the object of interests, by combining a user's mobile setting (location, time, and resource) with his or her personal interests	Personalized service based on known user profile (content and layout configuration without a need of log-in registration)
Communication	Targeted advertising suitable at the point-of-purchase	Customer feedback in multiple-answer or multimedia formats
Connection	Adaptive map that shows the information about nearby stores	The icon that gives a link to the starting page with one-click of 'cancel' button
Commerce	Insertion of authentication into mobile phones	One-click checkout process made available by storing a consumer's address, payment method, preferred delivery options

Table 2 : The Seven Design Elements of the M-Commerce Customer Interface (Lee &Benbasat, 2003)

Source: Ahmad and Ibrahim, 2017

Table 2 shows the 7Cs' design elements of m-commerce customer interface (Lee and Benbasat, 2003) as cited by Ahmad and Ibrahim (2017). These elements are specifically used in m-commerce transactions as an efficient means of retaining and increasing the number of buyers, which ultimately generate more revenue for the online market. The elements include communication, content, customization, commerce, context, connection, and community. These elements have been identified in the analysis conducted by Ahmad and Ibrahim (2017) and had provided two viewpoints, i.e. mobile settings (for the advantage of the customer's needs) and mobile device settings (for mobile device limitation). According to them, these findings can be used for mobile application design since the features as mentioned in Table 2 are also applicable to mobile technology.

The increasing numbers of studies in digital technology support the assumption that mobile apps are now a tool for economic and business activities due to being convenient, hassle-free, user-friendly, and accessible.

With the emergence of new mobile technologies, particularly, online shopping, the researchers have opted to develop the Market Mobile Application not only to increase online sales or e-commerce in the Philippines but also to promote local products, as well as to provide the primary sector with alternative means of ensuring a stable income even if they live in remote areas.

Figure 7: E-Commerce Activities in the Philippines (Percentage of Internet Users Aged 16 to 64)

Source: Datareportal, 2019

Figure 7 shows the Philippines' e-commerce activities for internet users aged 16 to 64 years old. It appears that 91% of them used the internet for online search for products or services, while 92% of the users visited an online retails store or website using any device; 75% of them purchased a product or service online using any device; 39% of the users purchased online using their laptop or desktop/computer. Finally, 62% of them purchased products using a mobile phone. These data show that mobile apps are now emerging for e-commerce. It was designed with multiple features and different functions to help people in their everyday activities, to interact with the world, to access information from a distance, to connect with social networks via Facebook or Twitter, and to search for whatever users need, among others (Islam, et al., 2010).

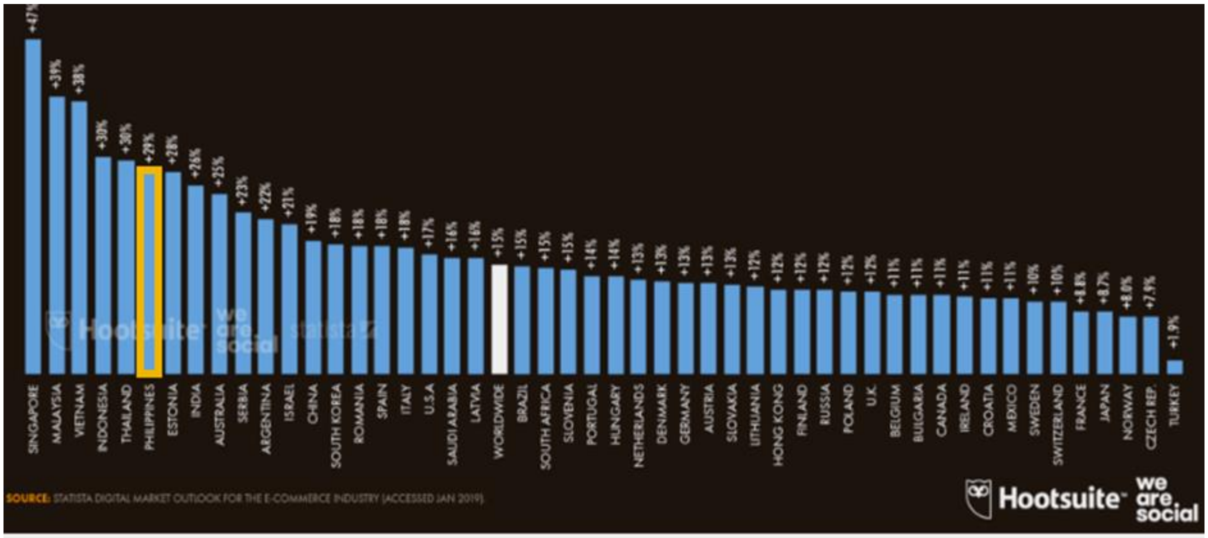


Figure 8: Annual Growth in Online Grocery Spend (Year-on-Year Change in the Value of Consumers' Online Grocery Purchases

Source: Datareportal, 2019

Online retail businesses, as illustrated in Figure 8, are growing in popularity in the Philippines receiving +29% of annual growth in online grocery spending compared to other countries. Singapore is the fastest-growing country in the world in terms of online shopping receiving +47% of its annual growth in online grocery spending, followed by Malaysia with +39%, then Vietnam with +38%, and so on.

Filipinos go online for approximately 10 hours a day, and 60% of them spend an average of 4 hours on Facebook, most of them engaged in purchasing and selling. The growth of online shopping in the Philippines continues to expand, resulting in the development of e-commerce digital platforms that sell a variety of products to millions of customers/buyers. However, while demand for online shopping is increasing, most products sold are items that can be purchased and distributed by grocery stores which have an impact on the livelihoods of the primary sectors living in remote communities, as local products are generally left behind.

On this premise, the researchers are extremely challenged in developing a market-oriented application for domestic products to identify, particularly, the needed requirements for primary sectors that do not have mobile phones compatible with any applications and no access to the internet.

The Department of Agriculture plays an important role in making this development a success. It is their mandate to promote agricultural development by providing a policy framework, public-private investment, and support services necessary for domestic and export business opportunities. It can also encourage public participation and involvement in sustainable agriculture, crafts, and fisheries through sectoral representation in policy-making, ensuring that departmental strategies, initiatives, and services are planned and implemented to meet their needs (DA RFO, n.d).

The challenge now for the public administration is how to help the primary sector that resides in remote areas, which are mostly illiterate and have limited skills to boost their economic activity. They remain ignorant due to a lack of awareness and sound information about prices, weather, government rules and regulations, news, training, credit markets, and market opportunities.

In many studies, a lack of awareness among local citizens has a significant impact on the growth of business operations, resulting in irrational decision-making (Molony, 2006). This simply means that information symmetry has been constantly a challenge, as most primary sectors have limited or no access to the needed information in a timely, accurate, consistent and reliable manner (Aker, 2011; Duncombe, 2012; Baumüller, 2012).

In the Duncombe (2012) study, it was noted that before the implementation of a mobile application such as Market Mobile application, it is necessary to understand first the needs and circumstances of the primary sectors. This includes the use of context-specific studies on access to mobile information services in agriculture, fisheries, forestry, mining, livestock, and craft industries. The level of the "digital divide" or the extent of the digital gap between the primary sector and technology should also be determined.

However, past research in most developing countries overlooked the challenges of the 'digital divide' in terms of basic personal skills (e.g. literacy or digital skills), motivational qualities, and the use of mobile phone information. In general, the knowledge needed by the primary sectors is divided into three categories: production system management, financial inclusion, and market access (Vodafone, 2005).

The researchers believe that, following the development of Market Mobile Application, integrating all the information needed by the primary sector, this application can ensure that they have a better opportunity to change and improve their lives (Cecchini & Scott, 2003).

In their study entitled "The Knowledge Mapping of Mobile Commerce Research: A Visual Analysis Based on I-Model", Du and Li (2019) illustrated the integrated framework of mobile commerce researches, as follows (Figure 9):

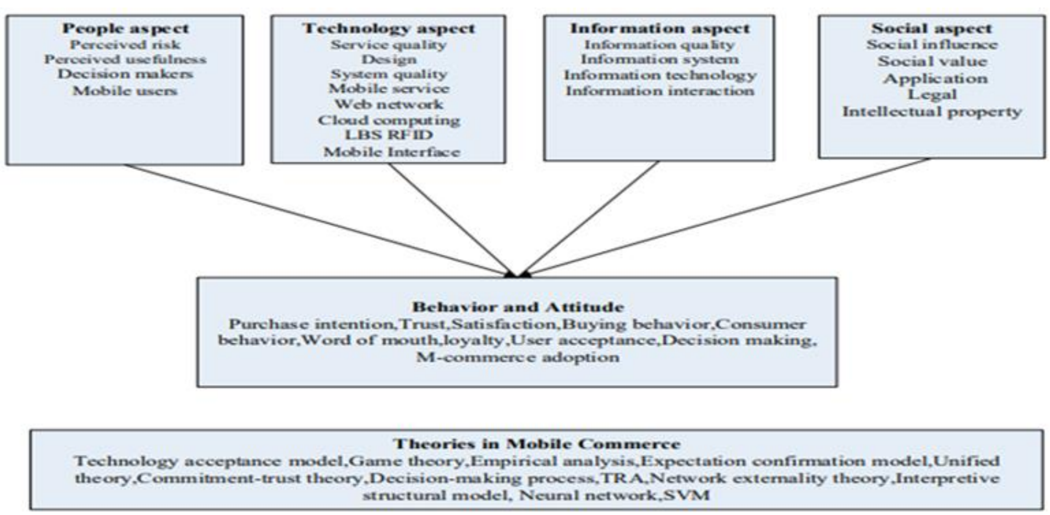
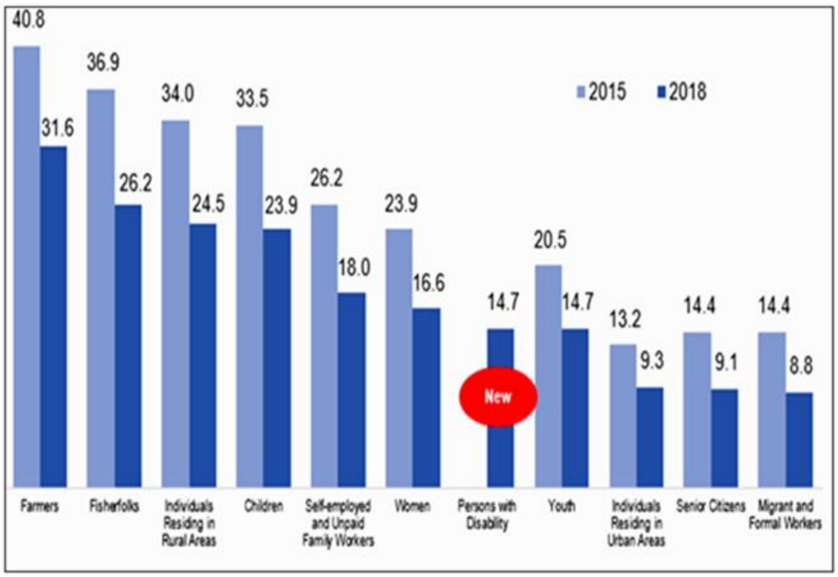


Figure 9. The Integrated Framework of Mobile Commerce Researches

This framework shows that the behavior and attitude of a person are greatly influenced by people, technology, information, and social aspects. These aspects show how a person perceived a certain situation, and how it influences their behavior, attitude, and decision. For example, if the primary sector, as a mobile user, does not have internet access despite having a compatible mobile phone, their ability to use it becomes useless, as they do not have internet access. As a result, their approach to technology is influenced by the fact that they do not have access to the Internet, not because they do not want to use it.

Primary Sector

As of June 3, 2020, the Philippine Statistics Authority (PSA) preliminarily estimated the incidence of poverty in 2018, for example, 10 of 14 basic sectors listed in Republic Act 8425, or the Social Reform and Poverty Alleviation Act, and those who are rural residents has been shown in Figure 10. The incidence of poverty among persons with disabilities (aged 15 and over) has also been produced and released, to wit:



Source: Merged data file of the 2015 Family Income and Expenditure Survey (FIES) and January 2016 Labor Force Survey (LFS) and preliminary merged data file of the 2018 FIES and January 2019 LFS, Philippine Statistics Authority

Note: r – revised; the 2015 estimates were revised/updated based on the following: a) rebasing of the Buyer Price Index (CPI) market basket of prices from 2006 to 2012; b) adoption of the 2015 Census of Population (PopCen) results for the weights in the merged FIES-LFS; and c) updated urban-rural classification of barangays based on the 2015 PopCen.

Figure 10. Poverty Incidence among the Basic Sectors (%):2015 and 2018

Figure 10 shows that the highest rates of poverty in key sectors in 2018 were 31.6, 26.2, 24.5, and 23.9 for farmers, fishermen, people living in rural areas, and children belonging to families below official poverty levels, respectively. These industries accounted for 40.8, 36.9, 34.0, and 33.5 in 2015, with the highest rates of poverty recorded (Mapa, 2020).

According to PSA, agriculture, hunting, forestry, and fisheries together accounted for 9.7% of the Philippines' GDP in the fourth quarter of 2016, decreasing by 1.1% annually. In recent years, extreme weather imbalances have resulted in unstable production. For instance, in 2015, with the major damage caused by Typhoon Lando, long dry spells have adversely affected the industry, which has recorded a production increase of just 0.11% since 2014. Production in crops and fisheries has dropped respectively by 1.95% and by 1.96% during 2015, while poultry and livestock segments increased by 5.7% and 3.8%, respectively. This climate change and the extended dry spell had a detrimental impact on crops in 2016, which caused a decrease of 1.3% in agriculture. As of 2015, increases in productivity in the livestock and poultry sectors were insufficient to compensate for losses in agricultural production and fisheries (Oxford). This issue could have been avoided if the primary industries were well informed and properly guided by the government.

Slow economic activity and poor performance in the Philippine agricultural sector have resulted in an increase or high incidence of poverty in the sector (Figure 10). The lack of knowledge of government initiatives, schemes, programs, and policies of the primary sector affects their effectiveness and their level of confidence in the government. The primary sector expects the government to help them protect their work especially during climate change, particularly in times of typhoons and drought, but due to lack of information and the distance, this climate change has significantly affected them.

Government-wide policies, initiatives, and the emergence of the latest digital mobile app will pave the way for the strengthening of the competitiveness and production of the primary sector in the future. These initiatives should be sustained by the government.

Market Mobile Application

Market Mobile application is the easiest and most accessible means for local communities, particularly the primary sector for convenient sales transactions by transforming the activities from traditional paper-based or wired Internet services to wireless Internet services (Karetsos, et al., 2014) to mobile software/application.

Mobile applications are also classified as "Self-Service Technology" devices that enable users to participate in different components of provider-buyer experiences, such as knowledge quest, cost scanners, and actual transactions (Kaushik and Rahman, 2015). Buyers invest about 25% of their real-time on shopping applications and concentrate more time on simple and quick texting, sports, videos, shows, and news/information applications (ComScore, 2014).

Domestic products of the Philippines

The Philippines' major agricultural crops are rice, corn, coconut, sugarcane, banana, cassava, pineapple, and vegetables. Cow, horse, carabao, goat, and dairy products are the major livestock products, while chicken and duck are the top products for poultry. The Philippines has wide coastal areas and an array of species of fish such as tuna and tuna-like varieties, sardines, anchovy, round-scads, and slip-mouth (Philippine Consulate General, Canada).

Almost every region in the Philippines produces grains of rice, corn, and coconut. That alone, when sold directly from the primary sector to buyers (Ps2C), will make their lives sufficient for them to live above the poverty line.

RESEARCH METHODOLOGY

The study adopts a Mixed Research Methodology. A quantitative method for collecting data from the result of a structured survey questionnaire using the Four Point Likert Scale. It would be measured using a simple percentage to get the mean and the median to have the verbal interpretation.

Figure 11. Triangulation Analysis

Qualitative method of the Scoping Review to identify gaps and/or priorities of the primary sector based on the results of several studies.

Both the survey findings and the scoping review will be synthesized and analyzed using the Triangulation Analysis (Figure 11) which is "used as an approach to strengthen the findings and build trust in the results of the study" (Bryman, 2004).

Data were collected online through Google forms in June 2020. Respondents were provided with links of the questionnaire through their email address and FB Messenger. The survey was participated by 284 primary sector-respondents and 275 buyer-respondents from different places in the Philippines. However, the survey is not, in any manner, a comprehensive representation of respondents' perception, as there is no equal representation of the primary sector due to mobility restrictions based on community quarantine in the Philippines.

RESULTS AND DISCUSSION

Demographic data of respondents are summarized as follows:

1. Sex

A. Primary Sector		
Sex	Ferquency	Percentage
Male	141	57.3
Female	105	42.7
Prefer not to say	0	0
Total	246	100
B. Consumer		
Sex	Ferquency	Percentage
Male	121	53.8
Female	98	43.55
Prefer not to say	6	2.65
Total	225	100

Table 3. Frequency and Percentage Distribution of the Respondents in terms of Sex

Table 3 shows the frequency and percentage of respondents by sex. The data on sex has been collected to know who used a mobile phone when it comes to selling and buying. 42.7% or 105 of the respondent primary sectors are female, while 57.3% or 141 are males. On the other hand, 53.7% or 121 of the Respondent-Buyers are female, while 43.55% or 98 are males, and 2.67% preferred not to mention their gender. From this data, most respondents in the primary sector who use mobile phones for business or sales transactions are male, while most buyers who purchase online or from mobile phones are female.

2. Productive Sector/Categories

Type	Ferquency	Percentage
Farmers	71	28.86
Fishermen	79	32.11
Miners	28	11.39
Livestock breeder and animal raiser	31	12.6
Forestry	21	8.54
Other	16	6.5
Total	246	100

Table 4. Frequency and Percentage Distribution of the Respondents in terms of Types of Primary Sector

Table 4 shows the frequency and percentage of respondents by occupation or type of primary sector. Data on occupation have been collected to determine the percentage of representation per occupation of the primary sector. 28.86% or 71 respondents are farmers, 32.22% or 79 respondents are fishermen, 11.39% or 28 respondents are miners, while 12.6% or 31 respondents are livestock breeders and animal raisers. There are also 8.54% or 21 respondents whose occupation is forestry, while other primary occupations have 6.5% or 16 respondents.

3. Age

A. Primary Sector

Type	Ferquency	Percentage
16 years old to 25 years old	20	8.13
26 years old to 35 years old	18	7.32
36 years old to 45 years old	95	38.62
46 years old to 55 years old	93	37.8
56 years old or older	20	8.13
Total	246	100

B. Buyer

Type	Ferquency	Percentage
16 years old to 25 years old	78	34.67
26 years old to 35 years old	83	36.89
36 years old to 45 years old	38	16.89
46 years old to 55 years old	19	8.44
56 years old or older	7	3.11
Total	225	100

Table 5. Frequency and Percentage Distribution of the Respondents in terms Age

Table 5 shows the age-related frequency and percentage of respondents. For the primary sector, 38.62% or 95 of the respondents are between 36 and 45 years of age, while 37.8 % or 93 respondents are in the age group 46 to 55 years old. The age group 16 to 25 years old, and 56 or older have a similar number of respondents, receiving 8.13% or 20 of the respondents, separately. While the 26 to 35 years old age group has 7.32% or 18 respondents. Age was included in this study to know their readiness to adopt mobile phone technology as part of their business activities.

On the other hand, most of the respondents-buyers belong to the age group 26 to 35 years old with 36.89% or 83 respondents, while 34.67% or 78 respondents belong to the age group 16 to 25 years old. The age group 36 to 45 years old have 16.89% or 38 respondents, while the age group 46 to 55 years old have 8.44% or 19 respondents. Lastly, the age group 56 years old and older have 3.11% or 7 respondents.

This data shows that most of the primary sector belongs to the older age group (36 to 55 years of age), while buyers who are interested to purchase domestic products using a mobile app belongs to the younger age group (16 to 35 years old).

The Survey Questionnaire using the Four-point Likert Scale

Rating Scale	Range	Verbal Descriptions	Verbal Interpretations
4	3.51-4.0	Strongly Agree	The respondents are very much in favor of the statement in question.
3	2.51-3.5	Agree	The respondents are in favor of the statement in question.
2	1.51 – 2.5	Disagree	The respondents are not in favor the statement in question.
1	1.00-1.5	Strongly Disagree	The respondents were extremely disapproved of the statement in question.

Table 6. Verbal Interpretation using Four Point Likert Scale

Table 6 provides a verbal interpretation of the results of the survey questionnaire distributed to both the primary sector and buyer-respondents.

Please select the phrase / word that best suited to your situation	Mean	Median	VI	Rank
1. You have an android/iOS mobile phone or have the capacity to buy an android/iOS mobile phone	2.29	2	Disagree	13
2. You have an internet connection or wifi connection, which you can access at any time	2.23	2	Disagree	14
3. Transportation groups such as Grab, Lalamove, Bus Terminal, etc. are available for delivery in your area when you purchase online or want some goods to be delivered to someone else.	3.30	3	Agree	10
4. You can pay or receive payment for purchases of goods or products through PayMaya, Paypal, GCash, Western Union, Smart Padala, etc.	3.58	4	Strongly Agree	9
5. You can download any mobile apps such as the Market Mobile App for online shopping to your mobile device	2.97	3	Agree	12
6. Direct sales between local industries and consumers are convenient and easy	3.60	4	Strongly Agree	7
7. Direct sales transactions between the primary sector and customers of local/domestic products will motivate and strengthen local industries.	3.65	4	Strongly Agree	3
8. You are willing to sell or purchase domestic goods or locally produced goods on-line using a mobile phone app.	3.58	4	Strongly Agree	9
9. Using mobile software applications such as the Market Mobile App as an online sales platform is a competitive way of dealing with a dynamically changing environment	3.62	4	Strongly Agree	6
10. Mobile apps software is a sustainable strategy to help local industries increase their profits.	3.59	4	Strongly Agree	8
11. You expect the goods/products to be delivered to you to be shipped or received in good quality or of superior value.	3.67	4	Strongly Agree	2
12. There should be terms and conditions of purchase, that stipulate that if the product delivered to the consumer is damaged and if such damage is not the fault of the consumer, you expect it to be replaced or refunded.	3.27	3	Agree	11
13. You expect items or goods to be delivered to you in the same form, size and packaging as shown in the mobile app picture.	3.64	4	Strongly Agree	4
14. Strengthening the promotion and sale of locally produced materials and products produced in the Philippines will contribute to economic growth and sustainable natural resources.	3.67	4	Strongly Agree	2
15. Strengthening domestic sales will help foster partnership, cooperation and local community support.	3.58	4	Strongly Agree	9
16. The Market Mobile App, as a means of selling domestic goods and products, is a timely approach and an innovative strategy in the current situation.	3.63	4	Strongly Agree	5
17. You believe that using the Market Mobile App software will help local industries expand their businesses and increase their revenues.	3.74	4	Strongly Agree	1
TOTAL	3.38	3.59	Strongly Agree	

Table 7. Summary Result of the Responses of the Respondents (Primary Sector and Buyer) in the Survey Questionnaire

Table 7 provides a summary of the responses from both the primary sector and the buyer. Most of the questions received a verbal interpretation of "Strongly Agree". However, a verbal interpretation of "Disagree" was obtained from the respondents' assessment on access to mobile devices (Item No. 1) and the internet (Item No. 2). This will support the argument that because most of the primary sector live remotely in the provinces, there is limited or no access to the Internet at all. They use mobile phones for communication purposes only. These findings demonstrate the reality of the 'digital divide,' a term that is commonly used to describe inequalities between access to the Internet and the use of other telecommunications facilities (Curtin, 2001).

In Item No. 3, respondents offer a verbal interpretation of "Agree" on the question on using transport groups such as Grab, Lalamove, and others, as a delivery service for the purchased products, while in Item No. 4, they responded "Strongly Agree" to use online payment for their sales transactions such as online banking, online payment centers, and credit cards. The use of electronic payment would make it possible for the primary sector to optimize sales and access funds more effectively, resulting in cash flow and enhanced buyers' experiences and relationships (Mayanja, 2020).

As to how mobile applications will impact the livelihoods of the primary sector, the researchers found that it depends on the benefits received by the users and their expectations and experiences. This was clarified in the scoping of additional empirical results from different studies and publications. Item Nos. 6 and 7 relate to the personal perception of the respondents on a direct sales transaction, showing a verbal interpretation of "Strongly Agree"; while Item Nos. 8, 10, and 17 refers to the use of mobile applications in sales transactions, which also received a verbal interpretation of "Strongly Agree"; same response to Item Nos. 11 and 13 which pertains to individual expectations on the mobile application; while Item Nos. 9, 14, 15, and 16 received "Strongly Agree" responses from both respondents, showing the impact and benefits of mobile phone apps on the sales and distribution of domestic products. This was discussed in the scoping review of other relevant studies and publications.

Russell and Steele (2013) noted that the following crucial issues leading to a 'digital divide' should be resolved with the digital revolution: 1) lack of connectivity due to poor communication infrastructure; 2) lack of support for the acquisition, or purchase of devices, hardware, internet connections, and software; 3) lack of awareness, learning, and workshop on digital and technical skills; and 4) lack of exchange of communication services. The framework of 'digital divide' has been further classified into three social implications: the first relates to inequalities to access the information technology; the second level is inequalities in the ability to use IT; and the third level deals with inequalities in results (Wei et al., 2011).

Most respondents support the idea of innovation and development in mobile applications because it embodies a holistic approach for a stable source of livelihood (Duncombe, 2012; Svensson & Wamala, 2012). Although limited in number, the primary sector respondents were genuinely interested and confident that the Market Mobile Application could be used as a rural development tool.

Quantitative findings have shown that primary sectors understand the potential of the Market Mobile Application technology to support livelihood opportunities through continuous interaction, buyer links, cooperation, collaborative effort and partnership, and cost-effective transactions.

The creation of the Market Mobile application is a significant and revolutionary connection strategy between the primary sector and the buyer. Mobile phone users and the smartphone technology sector have increased significantly in recent years (Turban, et al. 2018). The current "new normal" situation facing the Philippines reflects a growing desire on the part of the buyers to have an easy, versatile, portable touch-point transactions that are usually based on actual or physical shopping in the marketplace. The buyers' increasing use of self-service technology, and the use of mobile applications, is a key element contributing to the growth of mobile commerce (Newman, et al., 2018).

Synthesis of the results using Triangular Analysis

The analysis was conducted by evaluating the data sources of this study, such as the responses of primary sector-respondents and buyers-respondents to the survey questionnaire, the mapping of the theoretical principles and framework used by several authors, as well as their results, and the gaps identified by the researchers in their studies.

The researchers mapped different studies to identify possible gaps as enumerated in Table 1 (Scoping Review on Different Studies for the Development of Market Mobile Application), and from this analysis, adopted some of the theories applicable to the study, such as the Expectancy-Value Theory of Motivation (Figure 1), showing that both the primary sector and the buyer are willing to use the application based on their perceived benefit. Another theory used in this study is the Digital Divide, as explained in Figure 2, which shows that the digital divide is caused by social, culture, approach, geographical, and economic factors.

The researches from previous studies as illustrated in Table 1 with the statistics report shown in Figures 7, 8, 10, and 12 will strengthen the findings of this study and will help to improve the Market Mobile Application to provide the primary sector with a platform for advertising and selling its products and to make full use of mobile technology. This will provide them with a tool to meet their needs, not just through sales transactions, but also in other areas. This mobile application also includes other features such as the everyday environmental condition, weather reports, news, government initiatives and projects, training opportunities, best practices, chatbox, government policy, rules and regulations, and legislation relevant to their sector. Based on their responses and other research findings as explained in Table 7 (the result of the survey questionnaire), all the requirements the primary sector needs were incorporated into the creation of the Market Mobile app as illustrated in Figure 3 (Glocalization of Mobile Application) and Figure 6 (Process Flow of Market Activities). The subject matter and the discussions presented above in this study were the results of the review of other studies as presented in Tables 1 and 2, Figure 4 and this application contained the relevant details required to validate the study outcome.

CONCLUSION

Many of the primary sectors are left behind in terms of technology and, due to poverty, many are illiterate. Consequently, the public administration can aim to resolve this "digital divide" or technological disparity by providing them training and education opportunities. It should also offer an alternative to buying government-sponsored smartphones to marketing their products. The study is intended to support the primary sectors to raise their profits by enabling them to become entrepreneurs by direct contact with their buyers and to boost their socioeconomic status.

The buyers, the public administration, and the primary sectors expect more from each other and, if that expectation is fulfilled, they will appreciate the development of the Market Mobile Application and will ultimately be encouraged to continue to utilize the services it provides. The primary sector is the main supplier of local goods in the Philippines; the buyer, who assists the primary sector by constantly buying locally-made products from them; and, the Public Administration, who will supply the primary sector with all forms of development resources, such as mobile technology. This method supports the Expectancy Value Theory of Motivation, that people's expectations and sense of worth influence their actions and behavior.

The primary sector is the source of all raw materials for the creation of new products such as food, clothing, vehicles, shelters, and other needed products. The public administration should therefore help them to strengthen their industry by providing them with their needs, the opportunity to learn the technology, and the information needed to improve their sales activities. Ignoring their needs may lead to the depletion of the potential primary sector and their children. As noted, due to poverty, most of their children go to foreign lands in search of a better way to change their lives. If this is the case, the sector will be unmanned and the economic condition of the country will be adversely affected.

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